

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 29

- GEOMORPHIC PROCESSES**
 - **MASS MOVEMENTS**

GEOMORPHIC PROCESS

DENUDATION

DENUDATIONAL PROCESSES

PROCESS

Weathering

Mass
Movements

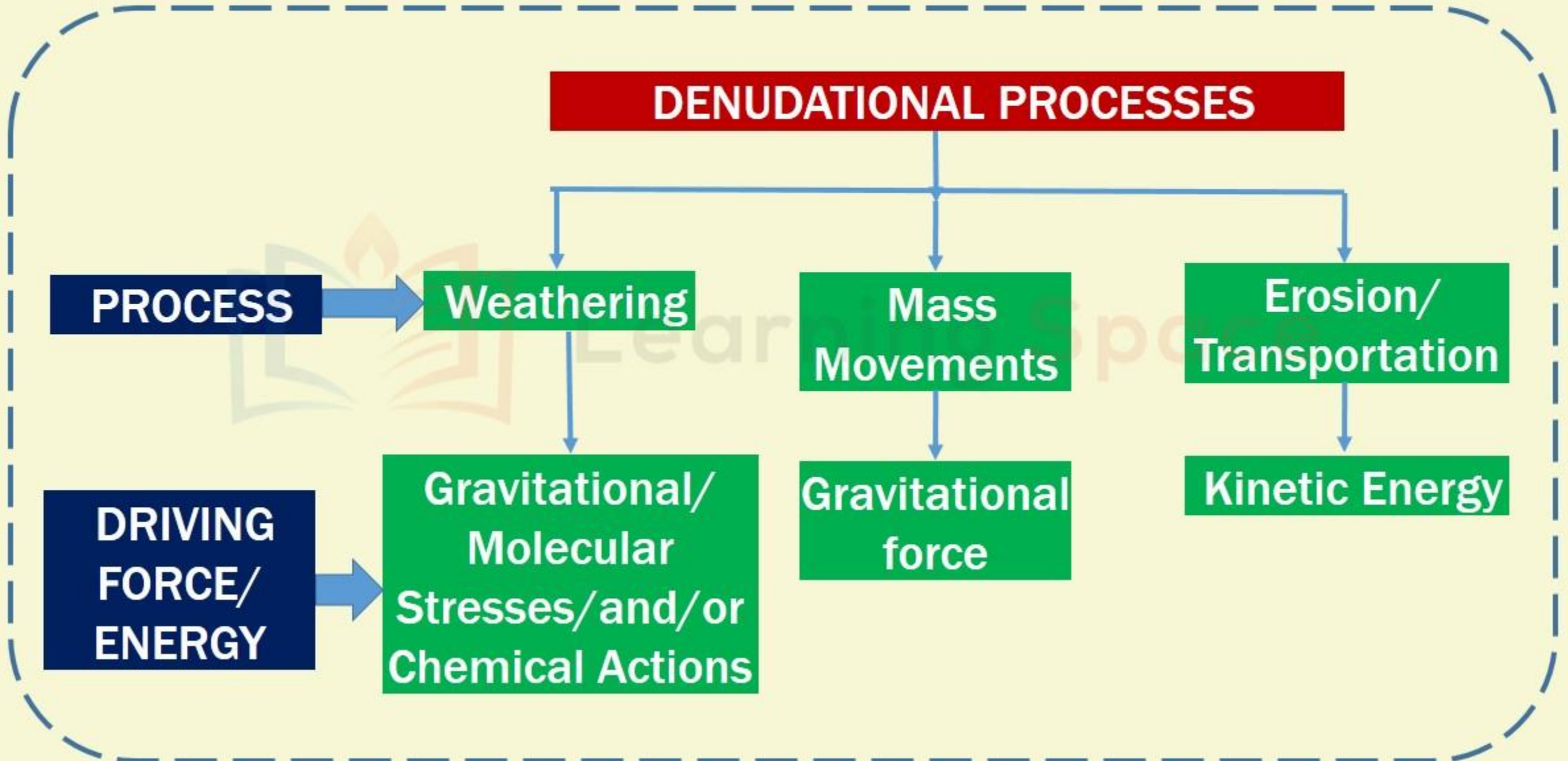
Erosion/
Transportation

DRIVING
FORCE/
ENERGY

Gravitational/
Molecular
Stresses/and/or
Chemical Actions

Gravitational
force

Kinetic Energy



MASS MOVEMENT



1 Mass movement is the movement of weathered materials down a slope due to gravitational forces.

2 The movement may be gradual or sudden depending on

- the gradient of the slope,
- the weight of the weathered debris and
- whether there is any lubricating moisture supplied by rain-water.

MASS MOVEMENT

1

SOIL CREEP



1

This is **slow, gradual but more or less continuous** movement of soil down hill slopes.

2

The movement is not very noticeable, especially when the slope is fairly gentle or when the soil is well-covered with grass or other vegetation.

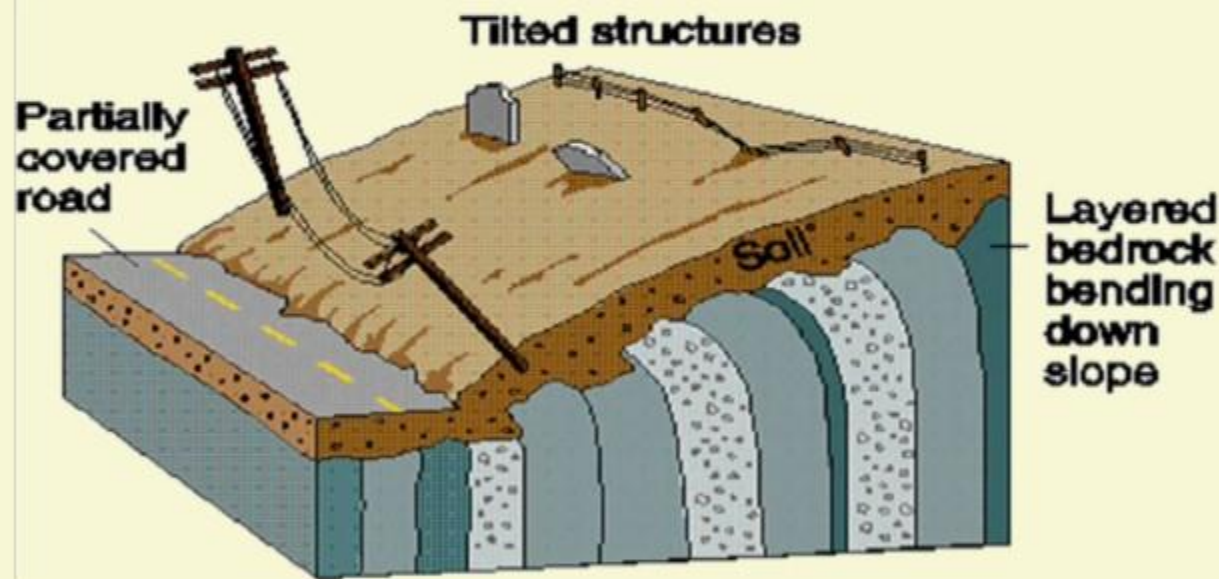
3

Soil creep is **most common in damp soils where the water acts as a lubricant** so that individual soil particles move over each other and over the underlying rock.

MASS MOVEMENT

1

SOIL CREEP



4

It is also found where continuous trampling by animals grazing on the slopes sets up vibrations which loosen the soil and cause it to move.

5

Though the movement is slow and cannot readily be seen in action, the gradual movement tilts trees, fences, posts and so on which are rooted in the soil.

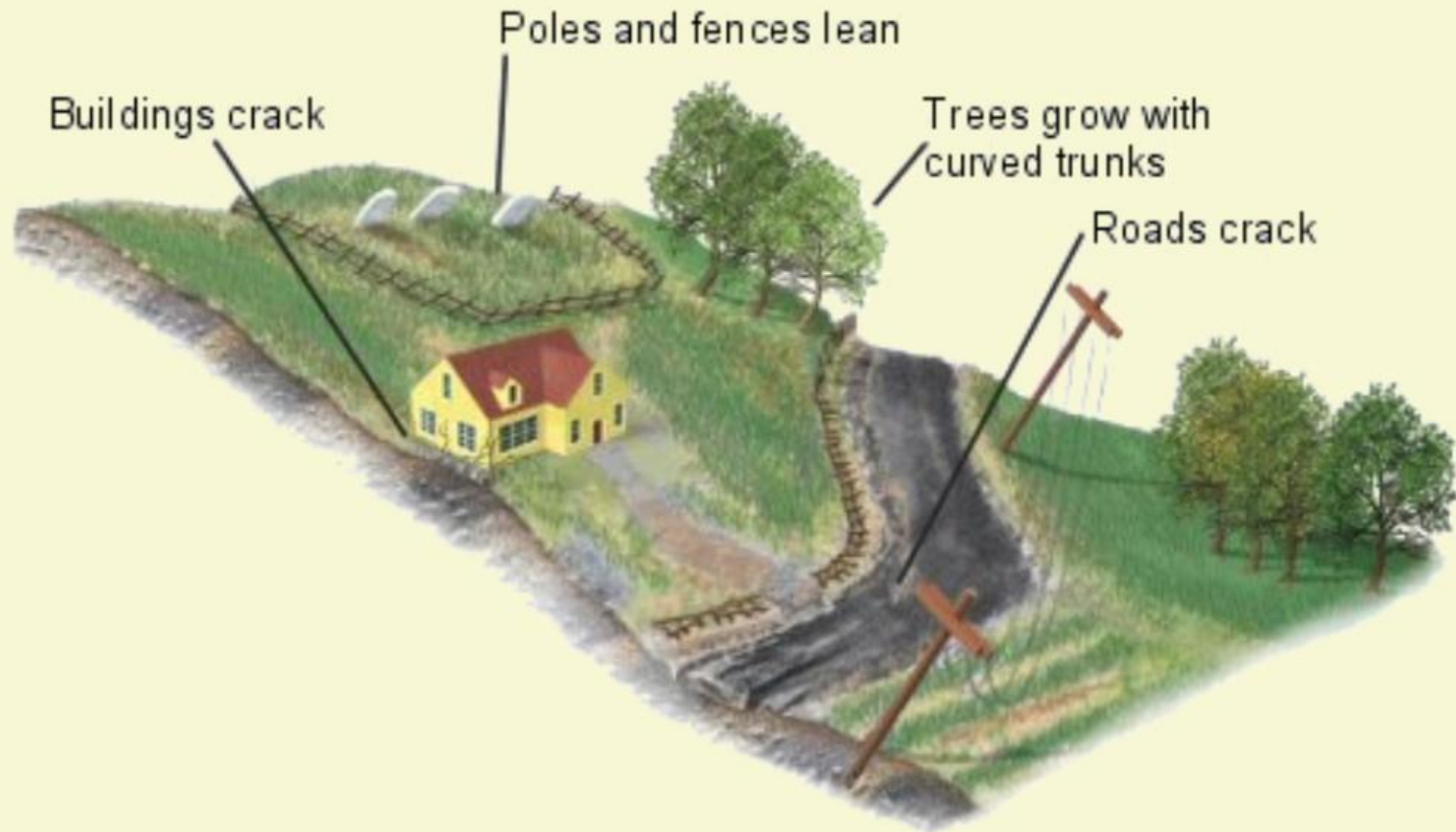
6

The soil is also seen to accumulate at the foot of slope or behind obstacles such as walls, which may eventually be burst by the weight of soil above.

MASS MOVEMENT

1

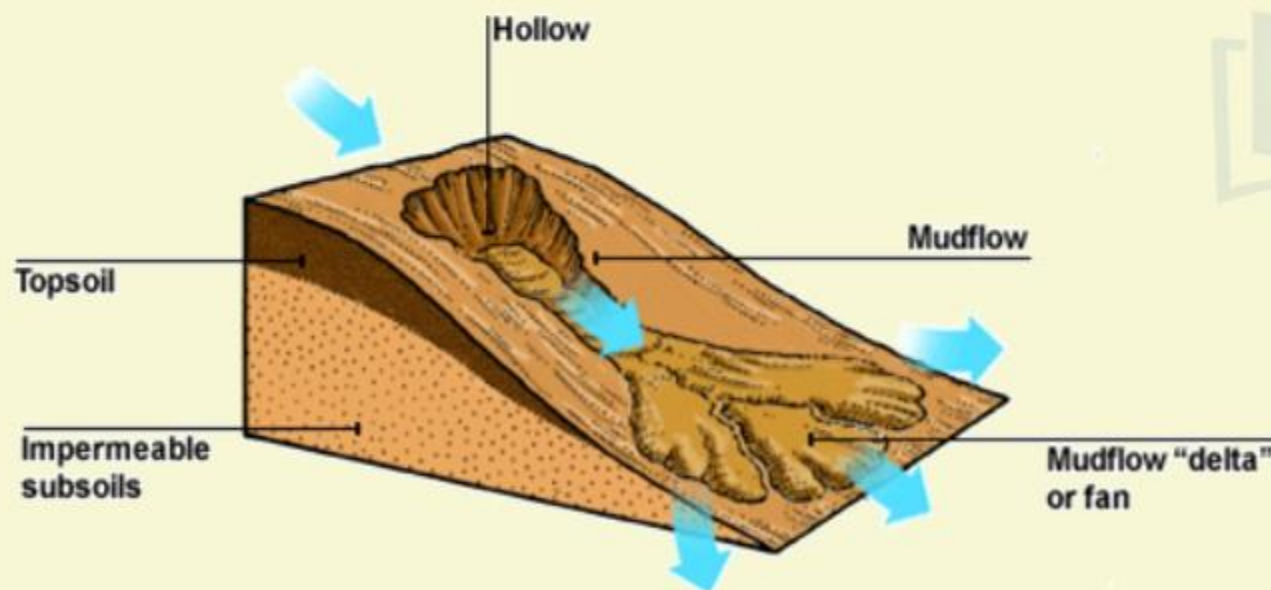
SOIL CREEP



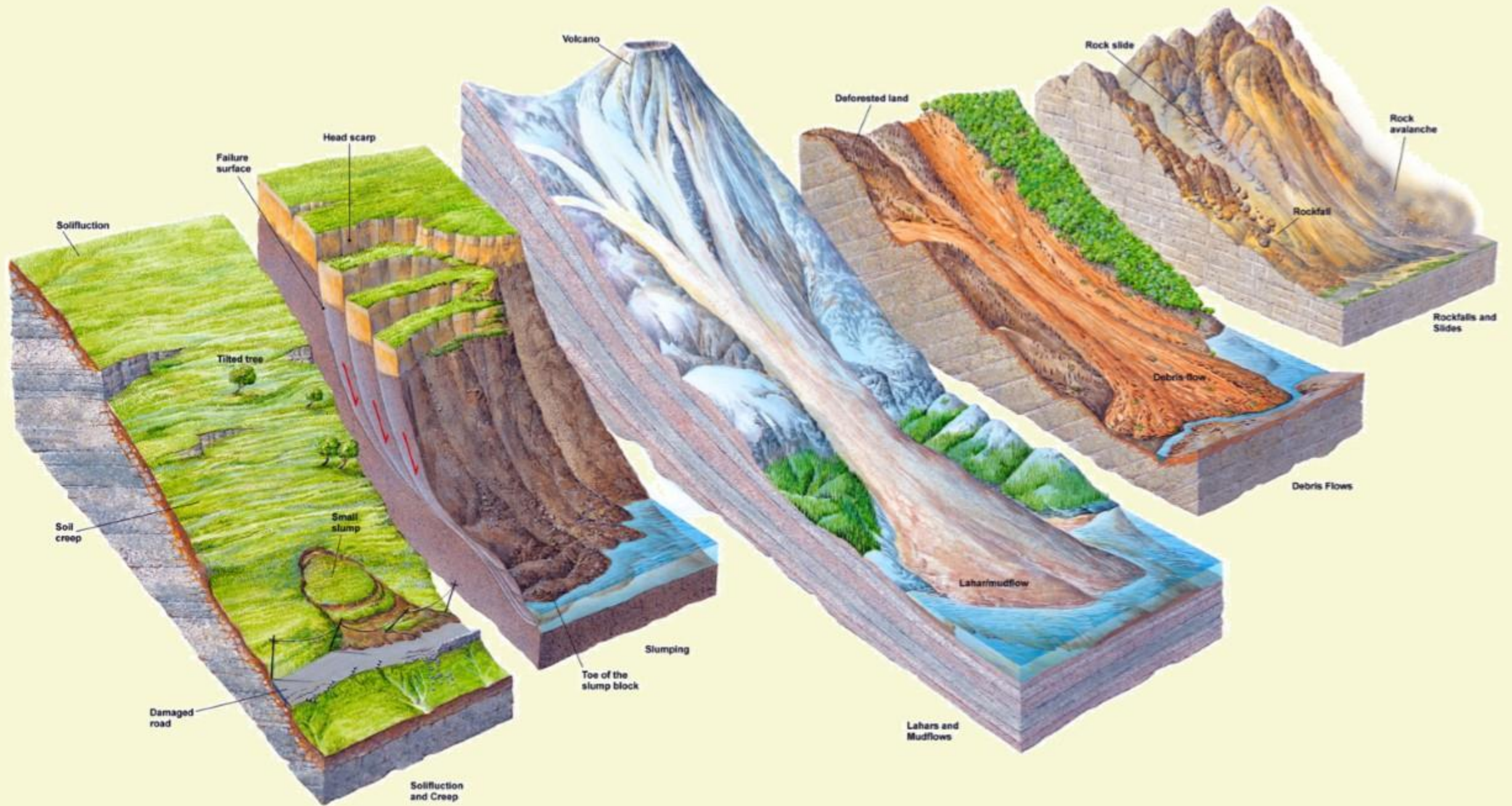
MASS MOVEMENT

2 SOIL FLOW (SOLIFLUCTION)

- 1 When the soil is completely saturated with water the individual particles are almost suspended in the water and move easily over one another and over the under-lying rock.



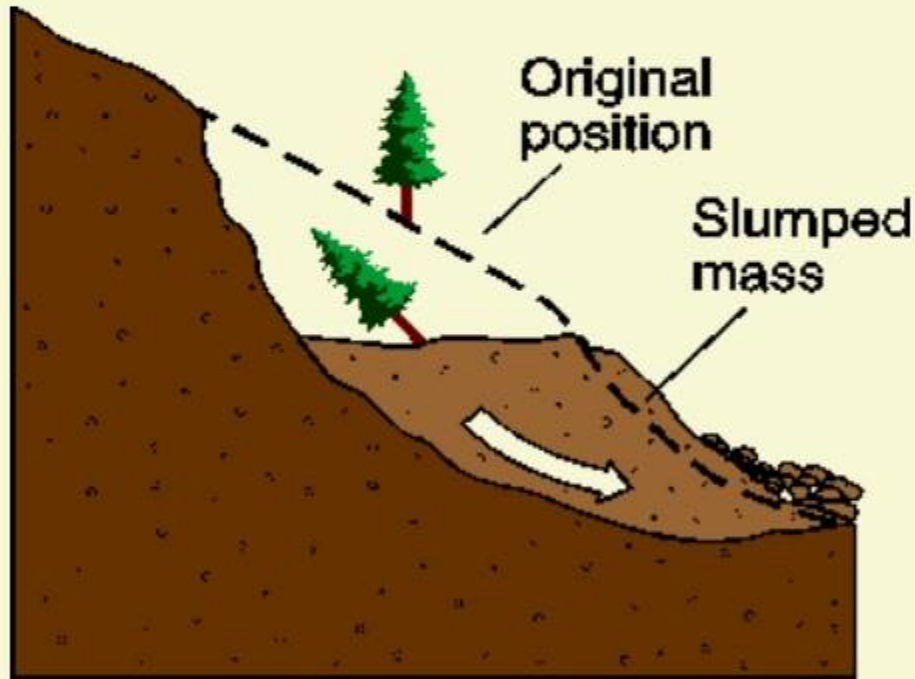
- 2 The soil acts like a liquid and a soil-flow or mud-flow occurs.
- 3 In arid regions a mantle of weathered debris may become saturated with rain-water after a storm and flow downslope as a semi-liquid mass.
- 4 In temperate and tundra regions, soil flow occur when the surface layers of frozen ground thaw in spring.
- 5 Soil and rock debris, lubricated by the melt-water, flow easily over the underlying frozen subsoil.



MASS MOVEMENT

3

LANDSLIDES (SLUMPING OR SLIDING)



1

These are **very rapid kinds of movement** and occur when a large mass of soil or rock falls suddenly.

2

Land-slides usually occur on steep slopes such as in mountainous areas, on cliffs or **where man has artificially steepened slopes**, for example, in road or rail cutting.

3

Landslides may be caused because a **steep slope is undercut by a river** or the sea so that it falls by gravity.

MASS MOVEMENT

3

LANDSLIDES (SLUMPING OR SLIDING)



Landslides caused by undercutting of river or the sea so that it falls by gravity.

MASS MOVEMENT

3

LANDSLIDES (SLUMPING OR SLIDING)



4 Earthquakes or volcanic disturbances may loosen rocks and start off a landslide.

5 Man-made steepening both undercuts the slope and sets up vibrations which may loosen rocks or soil.

6 Often landslides are caused by the lubricating action of rain-water. Water may collect in joints or bedding planes in rocks so that one layer slides over another, especially areas of tilted strata.

MASS MOVEMENT

3

LANDSLIDES (SLUMPING OR SLIDING)



7

The layer lying beneath provides a smooth slippery surface over which the upper layers easily slide.

8

Man often enhances the possibility of **landslides** by clearing natural vegetation for agriculture or housing.

9

Removal of the plant cover allows more water to penetrate the soil and rocks.

10

Extensive landslides, whether natural or man-induced, can have disastrous consequences, burying villages, railway lines or people.

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